1. Clinical Question & Core Hypothesis

Clinical Question

Ki-67 (MKI67) marks high proliferation and is often interpreted as predicting greater chemosensitivity, yet many highly proliferative breast tumors fail to show expected therapy responses. Why does high proliferation not reliably translate to improved chemotherapy response in some tumors? 2) Tiering / paths used: Because no Ontologist tiers were provided, I triaged the evidence myself: Tier 1 (core) — a proliferation→therapy-response axis (MKI67) and an extracellular-matrix/desmoplasia barrier axis (COL1A1) that can impede drug access and provide pro-survival signaling. I built the hypothesis on these two complementary mechanistic paths (proliferation-driven chemosensitivity vs. ECM-driven chemoprotection). 3) Tier 2 supporting evidence: ECM core matrisome genes (FN1, COL3A1) and stromal-rich histology (high TSR) provide biological plausibility that desmoplastic stroma reduces drug delivery, alters interstitial pressure, and activates survival pathways that blunt cytotoxic efficacy. 4) Why the endpoint: I chose a therapy_response proxy because the central puzzle is about why proliferation (MKI67) does not always predict treatment responsiveness; the available TCGA-BRCA context and the user's prompt specifically framed MKI67→therapy links. 5) What I excluded and why: I excluded immune-centric imaging features (TIL or TLS) and broad multi-gene signatures to keep the mechanistic hypothesis sharply focused on the interaction between proliferation and stromal ECM. Other candidate genes from broader pathways (e.g., transporters or TGF pathway nodes) were deliberately relegated to secondary modifiers to avoid overloading the core claim?

Working Hypothesis

High MKI67 expression in breast tumors predicts greater chemotherapy sensitivity only in contexts with low stromal-to-tumor ratio (TSR). In tumors with high TSR and COL1A1 expression, desmoplastic extracellular matrix impedes drug delivery and activates survival signaling, decoupling proliferation from chemotherapy response and reducing therapy efficacy; these tumors may benefit from combining cytotoxic therapy with stromal-targeting strategies.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	MKI67, COL1A1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MKI67: MKI67) marks high proliferation and is often interpreted as predicting greater chemosensitivity, yet many highly proliferative breast tumors fail to show expected therapy responses.

COL1A1: COL1A1) that can impede drug access and provide pro-survival signaling.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	FN1	Supporting modifier from knowledge graph
Secondary	COL3A1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: is a common predictor of chemotherapy response, but desmoplastic stroma can create physical and signaling barriers that negate proliferation-driven chemosensitivity. COL1A1-driven ECM density and related components (FN1, COL3A

Prediction 2: contribute to higher TSR, increasing interstitial pressure and limiting drug penetration while promoting survival pathways via integrin/FAK-YAP signaling.

Prediction 3: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- FN1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- COL3A1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.